

05 — Database & ER Diagram

The canonical schema is [backend/prisma/schema.prisma](#) (~1180 lines). Migrations live in [backend/prisma/migrations](#). PostgreSQL 14 on Azure Flexible Server.

Reading guide. The full ER diagram below is split by domain so it stays legible. A consolidated diagram follows.

1. Domains overview

Domain	Tables (logical names)
Identity	<code>users</code> , <code>refresh_tokens</code> , <code>password_reset_tokens</code> , <code>email_verification_tokens</code>
Customer data	<code>addresses</code> , <code>saved_payment_methods</code> , <code>favorites</code>
Loyalty	<code>loyalty_wallets</code> , <code>loyalty_transactions</code> , <code>loyalty_page_config</code>
Catalog core	<code>products</code> , <code>categories</code> , <code>subcategories</code> , <code>brands</code> , <code>designers</code> , <code>countries</code>
Catalog M:N	<code>product_categories</code> , <code>product_subcategories</code>
Catalog enrichment	<code>product_pricing</code> , <code>product_dimensions</code> , <code>product_packaging</code> , <code>product_images</code> , <code>product_specifications</code> , <code>product_overrides</code>
Variants	<code>product_groups</code> , <code>variant_attributes</code> , <code>product_variants</code>
Cart	<code>carts</code> , <code>cart_items</code>
Orders	<code>orders</code> , <code>order_items</code> , <code>order_status_history</code> , <code>order_status_master</code> , <code>invoices</code>
Returns	<code>returns</code> , <code>return_items</code>
Stock	<code>stock_movements</code>
Marketing	<code>promo_codes</code> , <code>announcements</code>
CMS storefront	<code>banners</code> , <code>landing_pages</code> , <code>landing_sections</code> , <code>navigation_menus</code> , <code>navigation_menu_items</code> , <code>product_collections</code> , <code>product_collection_items</code> , <code>home_page_config</code> , <code>home_page_sections</code> , <code>category_landing_page_config</code> , <code>category_landing_sections</code> , <code>site_settings</code>
Media	<code>media_assets</code>
B2B	<code>bulk_order_requests</code> , <code>bulk_order_items</code>

Domain	Tables (logical names)
Webhooks	<code>stripe_events</code>

2. Identity & customer

```

erDiagram
    USER ||--o{ REFRESH_TOKEN : has
    USER ||--o{ PASSWORD_RESET_TOKEN : has
    USER ||--o{ EMAIL_VERIFICATION_TOKEN : has
    USER ||--o{ ADDRESS : has
    USER ||--o{ SAVED_PAYMENT_METHOD : has
    USER ||--o{ FAVORITE : marks
    USER ||--|| LOYALTY_WALLET : has
    LOYALTY_WALLET ||--o{ LOYALTY_TRANSACTION : posts
    USER ||--o{ ORDER : places
    USER ||--o{ RETURN : files

    USER {
        string id PK
        string email UK
        string passwordHash
        enum role "CUSTOMER ADMIN SUPER_ADMIN"
        datetime emailVerifiedAt
        datetime deletedAt
    }
    REFRESH_TOKEN {
        string id PK
        string userId FK
        string tokenHash
        datetime expiresAt
        datetime revokedAt
        string replacedByTokenId
    }
    ADDRESS {
        string id PK
        string userId FK
        string line1
        string city
        string country
        bool isDefault
    }
    LOYALTY_WALLET {
        string id PK
        string userId FK,UK
        decimal balanceAed
    }
    LOYALTY_TRANSACTION {
        string id PK
        string walletId FK
        enum type "EARNED REDEEMED ADJUSTMENT EXPIRED"
        decimal amountAed
        string orderId
    }
    FAVORITE {
        string id PK

```

```
string userId FK
int productId FK
}
```

3. Catalog

```

erDiagram
    CATEGORY ||--o{ SUBCATEGORY : groups
    CATEGORY ||--o{ PRODUCT : "primary FK"
    SUBCATEGORY ||--o{ PRODUCT : "primary FK"
    BRAND ||--o{ PRODUCT : has
    DESIGNER ||--o{ PRODUCT : has
    COUNTRY ||--o{ PRODUCT : "country of origin"

    PRODUCT ||--o{ PRODUCT_CATEGORY : "M:N"
    CATEGORY ||--o{ PRODUCT_CATEGORY : "M:N"
    PRODUCT ||--o{ PRODUCT_SUBCATEGORY : "M:N"
    SUBCATEGORY ||--o{ PRODUCT_SUBCATEGORY : "M:N"

    PRODUCT ||--|| PRODUCT_PRICING : has
    PRODUCT ||--|| PRODUCT_DIMENSION : has
    PRODUCT ||--|| PRODUCT_PACKAGING : has
    PRODUCT ||--|| PRODUCT_OVERRIDE : has
    PRODUCT ||--o{ PRODUCT_IMAGE : has
    PRODUCT ||--o{ PRODUCT_SPECIFICATION : has
    PRODUCT ||--o{ VARIANT_ATTRIBUTE : has
    PRODUCT ||--o{ PRODUCT_VARIANTS : has
    PRODUCT_GROUP ||--o{ PRODUCT : siblings

    PRODUCT {
        int id PK
        string sku UK
        string slug UK
        string productName
        int categoryId FK
        int brandId FK
        string productGroupId FK
        int stockQty
        int reservedQty
        bool isActive
        bool isFeatured
        datetime deletedAt
    }
    PRODUCT_PRICING {
        int productId FK,UK
        decimal price_excl_vat_aed
        decimal price_incl_vat_aed
        decimal vatRate "default 0.05"
        bool isCurrent
    }
    PRODUCT_GROUP {
        string id PK
        string slug UK
        json variantAxes
    }
    VARIANT_ATTRIBUTE {

```

```
string id PK
int productId FK
string key
string value
string colorHex
}
```

4. Cart & orders

```

erDiagram
    USER ||--o{ CART : owns
    CART ||--o{ CART_ITEM : contains
    PRODUCT ||--o{ CART_ITEM : referenced
    USER ||--o{ ORDER : places
    ADDRESS ||--o{ ORDER : "shipping/billing"
    ORDER_STATUS_MASTER ||--o{ ORDER : status
    ORDER ||--o{ ORDER_ITEM : has
    ORDER ||--o{ ORDER_STATUS_HISTORY : timeline
    ORDER ||--o{ INVOICE : issues
    ORDER ||--o{ RETURN : may_have
    RETURN ||--o{ RETURN_ITEM : lines
    PROMO_CODE ||--o{ ORDER : applied

    CART {
        string id PK
        string userId FK
        string guestKey
    }
    CART_ITEM {
        string id PK
        string cartId FK
        enum type "PRODUCT"
        int productId FK
        int quantity
    }
    ORDER {
        string id PK
        string userId FK
        string orderNumber UK
        enum status
        enum paymentStatus
        enum paymentMethod
        enum shippingMethod
        decimal subtotalExclVat
        decimal vatAmount
        decimal vatRateSnapshot
        decimal shippingExclVat
        decimal discountExclVat
        decimal loyaltyRedeemAed
        decimal loyaltyEarnAed
        decimal totalInclVat
    }
    ORDER_ITEM {
        string id PK
        string orderId FK
        int productId
        string sku
        string name
        int quantity
    }

```

```

    decimal unitPriceExclVat
    decimal unitVatAmount
    decimal unitPriceInclVat
    decimal lineTotalInclVat
}
INVOICE {
    string id PK
    string orderId FK,UK
    string invoiceNumber UK
    decimal vatRateSnapshot
    string pdfUrl
}

```

5. Returns & stock

```

erDiagram
    ORDER ||--o{ RETURN : has
    RETURN ||--o{ RETURN_ITEM : lines
    USER ||--o{ RETURN : files
    PRODUCT ||--o{ STOCK_MOVEMENT : ledger

    RETURN {
        string id PK
        string returnNumber UK
        string orderId FK
        string userId FK
        enum status "REQUESTED APPROVED RECEIVED COMPLETED REJECTED"
        enum reason
        enum refundMethod "ORIGINAL_PAYMENT LOYALTY_AED STORE_CREDIT"
        decimal refundAmount
        bool stockRestored
    }
    STOCK_MOVEMENT {
        string id PK
        int productId FK
        int quantity
        enum type "IN OUT ADJUSTMENT RESERVATION RELEASE"
        string reference
    }

```

6. CMS / storefront

```

erDiagram
    MEDIA_ASSET ||--o{ BANNER : "image / mobile_image"
    BANNER ||--o{ LANDING_PAGE : hero
    LANDING_PAGE ||--o{ LANDING_SECTION : has
    NAVIGATION_MENU ||--o{ NAVIGATION_MENU_ITEM : has
    NAVIGATION_MENU_ITEM ||--o{ NAVIGATION_MENU_ITEM : children
    PRODUCT_COLLECTION ||--o{ PRODUCT_COLLECTION_ITEM : has
    HOME_PAGE_CONFIG ||--o{ HOME_PAGE_SECTION : has
    CATEGORY_LANDING_PAGE_CONFIG ||--o{ CATEGORY_LANDING_SECTION : has
    PROMO_CODE ||--o{ ANNOUNCEMENT : may_link

    BANNER {
        string id PK
        enum placement "HOME_HERO HOME_MID CATEGORY_HERO ..."
        string mediaAssetId FK
        string mobileMediaAssetId FK
        datetime startsAt
        datetime endsAt
    }
    HOME_PAGE_SECTION {
        string id PK
        string configId FK
        enum type "HERO BANNERS CATEGORIES COLLECTION BRANDS NEW_ARRIVALS BEST_SELLERS ..."
        int sortOrder
        json config
    }
    ANNOUNCEMENT {
        string id PK
        string text
        string promoCodeId FK
        datetime startsAt
        datetime expiresAt
    }

```

7. B2B & integrations

```
erDiagram
    BULK_ORDER_REQUEST ||--o{ BULK_ORDER_ITEM : lines
    STRIPE_EVENT ||--|| STRIPE_EVENT : "deduplicated by id"

    BULK_ORDER_REQUEST {
        string id PK
        int orderNumber UK
        string name
        string email
        string phone
        enum status "NEW CONTACTED QUOTED CONVERTED CLOSED"
    }
    STRIPE_EVENT {
        string id PK
        string type
        bool processed
        json payload
    }
```

8. Consolidated ER diagram

```

erDiagram
    USER ||--o{ ORDER : places
    USER ||--|| LOYALTY_WALLET : has
    LOYALTY_WALLET ||--o{ LOYALTY_TRANSACTION : posts
    USER ||--o{ ADDRESS : has
    USER ||--o{ FAVORITE : marks
    USER ||--o{ CART : owns
    CART ||--o{ CART_ITEM : contains
    PRODUCT ||--o{ CART_ITEM : in
    CATEGORY ||--o{ PRODUCT : primary
    BRAND ||--o{ PRODUCT : has
    DESIGNER ||--o{ PRODUCT : has
    COUNTRY ||--o{ PRODUCT : origin
    PRODUCT_GROUP ||--o{ PRODUCT : siblings
    PRODUCT ||--|| PRODUCT_PRICING : has
    PRODUCT ||--o{ PRODUCT_IMAGE : has
    PRODUCT ||--o{ STOCK_MOVEMENT : ledger
    ORDER ||--o{ ORDER_ITEM : has
    ORDER ||--o{ INVOICE : issues
    ORDER ||--o{ ORDER_STATUS_HISTORY : timeline
    ORDER ||--o{ RETURN : may_have
    RETURN ||--o{ RETURN_ITEM : lines
    MEDIA_ASSET ||--o{ BANNER : image
    BANNER ||--o{ LANDING_PAGE : hero
    LANDING_PAGE ||--o{ LANDING_SECTION : has
    HOME_PAGE_CONFIG ||--o{ HOME_PAGE_SECTION : has
    PRODUCT_COLLECTION ||--o{ PRODUCT_COLLECTION_ITEM : has
    NAVIGATION_MENU ||--o{ NAVIGATION_MENU_ITEM : has
    PROMO_CODE ||--o{ ANNOUNCEMENT : may_link
    BULK_ORDER_REQUEST ||--o{ BULK_ORDER_ITEM : lines

```

9. Enums (selected)

Enum	Values
UserRole	CUSTOMER , ADMIN , SUPER_ADMIN
OrderStatus	PENDING , PAYMENT_PENDING , PAID , PROCESSING , SHIPPED , DELIVERED , CANCELLED , REFUNDED
PaymentStatus	PENDING , AUTHORIZED , PAID , FAILED , REFUNDED
PaymentMethod	CREDIT_CARD , CASH_ON_DELIVERY , TABBY , TAMARA
ShippingMethod	STANDARD , EXPRESS
LoyaltyTransactionType	EARNED , REDEEMED , ADJUSTMENT , EXPIRED
CartItemType	PRODUCT

Enum	Values
ReturnStatus	REQUESTED , APPROVED , RECEIVED , COMPLETED , REJECTED
ReturnReason	DEFECTIVE , WRONG_ITEM , NOT_AS_DESCRIBED , CHANGE_OF_MIND , OTHER
RefundMethod	ORIGINAL_PAYMENT , LOYALTY_AED , STORE_CREDIT
StockMovementType	IN , OUT , ADJUSTMENT , RESERVATION , RELEASE
BannerPlacement	HOME_HERO , HOME_MID , CATEGORY_HERO , PRODUCT_BANNER , ...
HomeSectionType	HERO , BANNERS , CATEGORIES , COLLECTION , BRANDS , NEW_ARRIVALS , BEST_SELLERS , FEATURED , ANNOUNCEMENT
LandingSectionType	HERO , IMAGE_TEXT , PRODUCT_GRID , BANNER , RICH_TEXT , VIDEO , COLLECTION
CategoryLandingSectionType	HERO , SUBCATEGORIES , BANNER , PRODUCT_GRID , RICH_TEXT
ProductCollectionStrategy	MANUAL , AUTO
BulkOrderStatus	NEW , CONTACTED , QUOTED , CONVERTED , CLOSED
PromoType	PERCENT , FIXED , FREE_SHIPPING

10. Indexing summary (high-traffic)

Table	Index	Purpose
products	slug , sku , categoryId , brandId , isFeatured , isActive , deletedAt , productGroupId , createdAt	Listing/sort
product_pricing	productId	Join performance
orders	userId , status , paymentStatus , createdAt , orderNumber UK	Queue & history
order_items	orderId , productId	Reports
cart_items	cartId , productId	Cart read
refresh_tokens	userId , tokenHash UK, expiresAt	Auth
stripe_events	processed, createdAt	Reconciliation
stock_movements	productId , type , createdAt	Ledger
announcements	isActive , startsAt , expiresAt , sortOrder	Storefront
media_assets	folder , ownerType, ownerId , isDeleted	Library

11. Migrations & seeding

- Generate Prisma client: `npm run prisma:generate`
- Apply migrations (dev): `npm run prisma:migrate`
- Apply migrations (prod): `prisma migrate deploy`
- Seed: `npm run db:seed` → `prisma/seed.ts`
- Backups: workspace `backups/` plus `solo_ecommerce_pre_cleanup_20260208.dump` (pre-cleanup snapshot).
- Azure Postgres Flexible Server has automated daily backups (7-day retention by default; tune in [infra/main.bicep](#)).